

Leander Paul

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Dubai, United Arab Emirates

PROFESSIONAL SUMMARY

AI Engineer and Software Architect with 8+ years of experience, specialising in production LLM systems: agentic workflow orchestration with LangChain and LangGraph, retrieval-augmented generation, multi-provider model routing and context engineering at scale. Designed and shipped Novel Forge, an AI novel-generation platform built on 7 checkpointed LangGraph StateGraph workflows with hybrid pgvector retrieval and per-call cost telemetry. Background spans banking, IAM, fintech and AI startups, including production services at Emirates NBD and identity systems serving millions of users at Zoho.

PROFESSIONAL EXPERIENCE

Sphere IT Consultants (Client: Emirates NBD)

March 2024 - Present

Software Engineer

Dubai, United Arab Emirates

- Architected and built **9+ backend microservices** (Node.js, Express) on a team operating a platform of **30+ microservices** for retail mobile banking, serving millions of customers across the UAE.
- Designed a configurable multi-factor authentication system supporting OTP, IVR and Nafath (KSA national identity), with mid-flow validation checkpoints across multi-step authorization journeys.
- Engineered the end-to-end credit card application journey, integrating external credit scoring and employment verification services, plus the STP (Straight Through Processing) flow for new-to-bank onboarding with manual review stages.
- Built a custom automated testing suite **achieving 95% code coverage and reducing manual testing time by 50%**.
- Led Disaster Recovery environment setup and deployment planning; resolve production-critical incidents affecting live banking services.
- Technologies:** Java, Spring Boot, Node.js, Express, TypeScript, Microservices, REST APIs, PostgreSQL, Oracle.

WTV Global

March 2023 - September 2023

Full Stack Engineer

London, United Kingdom

- Designed the architecture for Capture, a carbon accounting platform, as **3 NestJS microservices communicating via RabbitMQ**, with independent service boundaries for auth, project permissions and carbon calculations.
- Built the full backend using NestJS and Fastify with MongoDB, and contributed to the React and Tailwind CSS frontend.
- Initiated end-to-end testing infrastructure with automated database seeding, **achieving a 30% reduction in system downtime**.
- Technologies:** NestJS, Fastify, React, RabbitMQ, Docker, MongoDB, Tailwind CSS, Stripe.

Zoho Corporation

January 2020 - August 2022

Software Engineer

Chennai, India

- Member of the Zoho Accounts IAM team, building identity and access management systems serving millions of users.
- Implemented SAML authentication and designed a hierarchy-based role and permission architecture, improving maintainability of the access-control model.
- Diagnosed and resolved a critical Redis bottleneck caused by TTL overhead, **reducing CPU usage by 60%** across the caching infrastructure.
- Built a proof-of-concept integrating Apache Directory Server with PostgreSQL to enable LDAP queries against the production database; built migration tooling for application-level sharded databases.
- Mentored 3 junior engineers and established internal best-practices documentation for the IAM team.
- Technologies:** Java, JavaScript, PostgreSQL, Redis, Apache Directory Server, SAML, LDAP.

PhospheneAI

April 2018 - January 2020

Full Stack Engineer & Team Lead

Chennai, India

- Founding engineering team member responsible for the architecture and delivery of PixelView, an AI image upscaling platform.
- Engineered a cost-optimized pipeline (client-side S3 uploads, Lambda triggers, SQS task queues, auto-scaling GPU servers idling after 5 minutes) **reducing infrastructure cost by 75%**.
- Led a team of 3 engineers, owning 90% of the codebase; deployed on AWS at **99% uptime** and **reduced frontend load time by 30%**.
- **Technologies:** Node.js, Next.js, React, Python, AWS (Lambda, SQS, EC2, S3), MongoDB, Stripe.

Freelance

April 2017 - March 2018

Software Engineer

Chennai, India

- Built and deployed 2 full-stack web applications: Bluebell (form management) and Jute Emporium (e-commerce), with Razorpay payments, order tracking and custom admin panels. **Technologies:** Node.js, MongoDB, AWS EC2.

PROJECTS

Novel Forge - AI Novel Generation Platform

AI platform generating full-length novels. 7 LangGraph StateGraph workflows checkpointed to PostgreSQL, so a crash mid-generation resumes at the last completed node instead of re-paying the model call. Generate-judge-repair loop: a judge model flags continuity contradictions and a repair chain applies find/replace patches only where the target text occurs exactly once. Model router across Anthropic, OpenAI, xAI and local Ollama (22 roles, 7 groups); structured output via 25 Zod schemas with a parse-repair ladder; hybrid retrieval over two pgvector indexes; token budgeting, prompt caching and per-call telemetry. Built and operated solo as part of a 4-product monorepo: 8 deployable apps, ~215K LOC TypeScript, 283 test suites.

LangChain, LangGraph, LlamaIndex, pgvector, PostgreSQL, TypeScript, Bun, Drizzle

Code Atlas - AI Codebase Assistant (Hybrid RAG)

AI codebase assistant combining semantic search, lexical search and symbol-graph navigation over a Tree-sitter AST parse. Tool-calling agents produce grounded, citation-backed responses, measured by an evaluation framework covering Recall@K, MRR, nDCG, citation-grounding verification and LLM-as-Judge scoring. 317 automated tests, full CI.

Python, FastAPI, LanceDB, SQLite FTS5, NetworkX, Tree-sitter, Ollama, Docker

EDUCATION

Master of Science in Computer Science

2022 - 2023

Queen Mary University of London

Distinction equivalent (4.0/4.0). Advanced Software Engineering, Distributed Systems, Data Systems, Algorithms, System Design.

Bachelor of Technology in Information Technology

2016 - 2020

St. Joseph's College of Engineering, Anna University

CGPA 7.5/10.

TECHNICAL SKILLS

AI and Machine Learning: LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), Vector Databases, pgvector, Embeddings, Prompt Engineering, Prompt Caching, LLM Evaluation, LLM-as-Judge, AI Agents, Tool Calling, Model Context Protocol (MCP), Structured Output, Model Routing, Ollama, Anthropic Claude, OpenAI, Context Engineering

Languages and Backend: TypeScript, JavaScript, Java, Python, SQL, Node.js, Bun, NestJS, Fastify, Express, Spring Boot, Microservices, REST APIs, Event-Driven Architecture, Domain-Driven Design

Frontend and Data: React, Next.js, Vue.js, Angular, TanStack Start, Tailwind CSS, PostgreSQL, pgvector, MongoDB, Redis, Drizzle ORM, SQLite, LanceDB

Cloud, DevOps and Identity: AWS (EC2, Lambda, S3, SQS), Docker, Kubernetes, ArgoCD, GitHub Actions, CI/CD, OAuth 2.0 / 2.1, OIDC, SAML, LDAP, RBAC, IAM, Multi-Factor Authentication

Practices: System Design, Software Architecture, SOLID, Distributed Systems, High Availability, Disaster Recovery, Test Automation, Mentoring